

# Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

## Executive Technology Brief

**Domain:** mechanical\_structural

**Technology Maturity:** ENGINEERING\_DEFINITION

**Transfer Posture:** Sponsored Validation

**Problem:** Documented recurring problem 'lithium-ion consumer-product battery fire': 5 of 50 CPSC recalls in the retrieved window involve batteries/chargers/power electronics. Recalled products: 'Cooluli Recalls 10-Liter and 15-Liter Minifridges Due to Fire and Burn Hazards'; 'Shengsen Prank Sound Makers Recalled Due to Risk of Serious Injury or Death from Battery Ingestion'; 'A2batt Recalls EEMB Lithium Coin Battery Chargers Due to Risk of Serious Injury or Death from Batter'. LIMITS STAMPED INLINE: a recall is an acknowledged product hazard, NOT an incidence rate; injury lists are associated complaints, not a census (Art. XXI.5 analog).

**Mechanism:** smart battery management systems that enhance resilience while maintaining energy density

**Architecture:** PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

**Governing Model:** Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

## Key Equations:

MEC-002: Section bending stress. Expression:  $\sigma = M * c / I$ . Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression:  $\sigma = E * \epsilon$ . Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression:  $P_{cr} = \pi^2 * E * I / (K * L)^2$ . Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL\_DERIVED. Engaged variables: NOT ESTABLISHED.

**Kill Condition:** killer experiment fails: falsification\_test\_from\_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING\_DEFINITION. No physical prototype. No clinical validation.

*Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.*